Night Shift

On-call agents that never solve the same incident twice, even when the other shift solved it.

ContextCon: Remember Everything · Theme 02 Agentic support: Never Solve It Twice

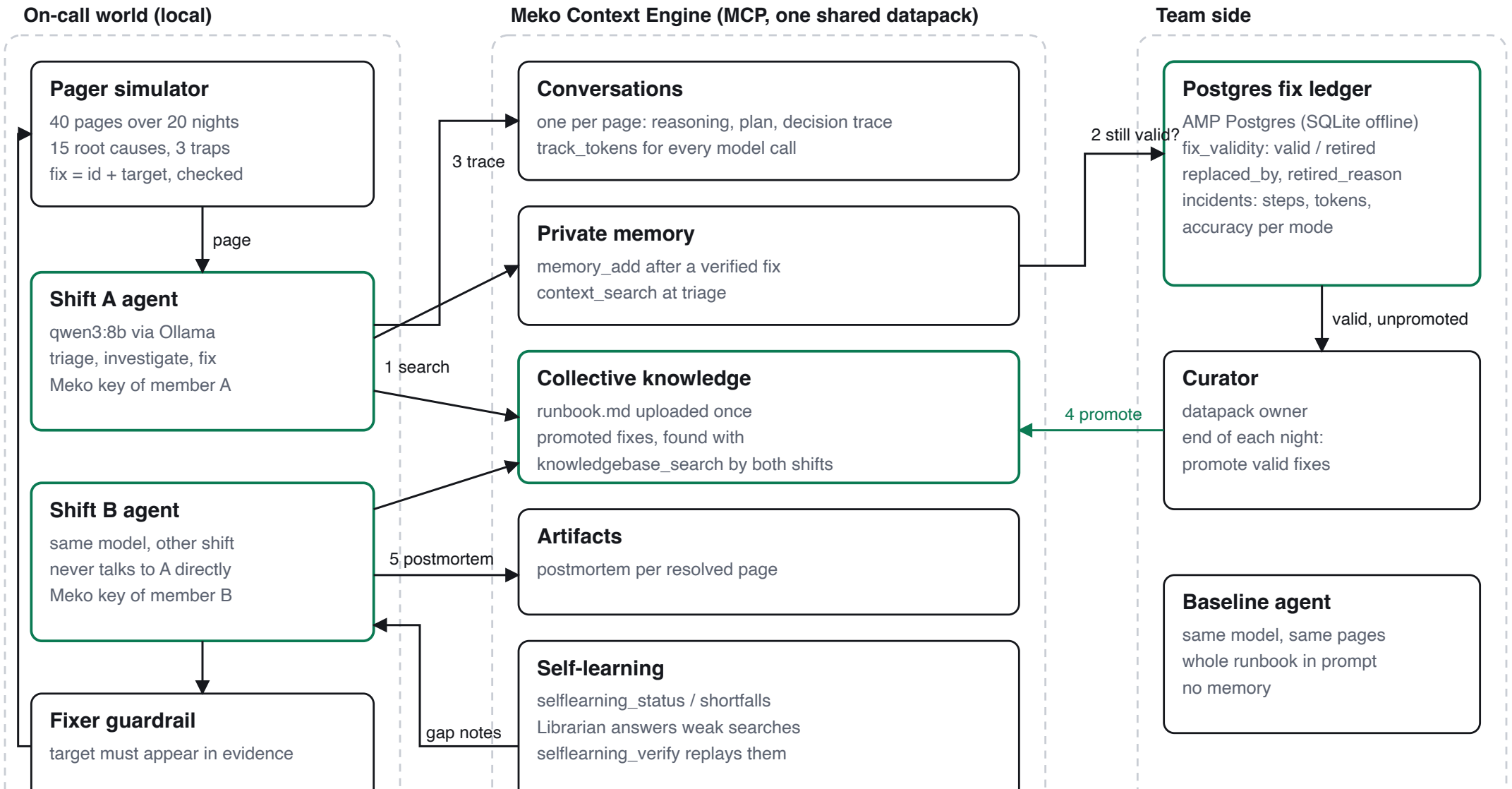
github.com/kangraemin/nightshift-meko

The same page gets solved twice

- Internal on-call teams split the night into shifts.
- The engineer who fixed the 2 a.m. page writes it down in their own notes, if at all.
- The next shift gets the same page and starts from the wiki again.
- The wiki is half right: it describes last year's causes, not this week's.
- Worse, a remembered fix can stop working. Applying it again wastes the first ten minutes of an outage.

Night Shift turns each verified fix into team memory, and keeps a ledger of which remembered fixes still work.

How it works


On-call world (local)
Pager simulator
40 pages over 20 nights
15 root causes, 3 traps
fix = id + target, checked
page
Shift A agent
qwen3:8b via Ollama
triage, investigate, fix
Meko key of member A
Shift B agent
same model, other shift
never talks to A directly
Meko key of member B
Fixer guardrail
target must appear in evidence
Meko Context Engine (MCP, one shared datapack)
3 trace
1 search
5 postmortem
gap notes
Conversations
one per page: reasoning, plan, decision trace
track_tokens for every model call
Private memory
memory_add after a verified fix
context_search at triage
Collective knowledge
runbook.md uploaded once
promoted fixes, found with
knowledgebase_search by both shifts
Artifacts
postmortem per resolved page
Self-learning
selflearning_status / shortfalls
Librarian answers weak searches
selflearning_verify replays them
Team side
2 still valid?
Postgres fix ledger
AMP Postgres (SQLite offline)
fix_validity: valid / retired
replaced_by, retired_reason
incidents: steps, tokens,
accuracy per mode
valid, unpromoted
Curator
datapack owner
end of each night:
promote valid fixes
4 promote
Baseline agent
same model, same pages
whole runbook in prompt
no memory

One page, end to end

- **Triage.** The shift searches its own memory and the team's collective knowledge in Meko.
- **Check.** Every hit is joined with the Postgres fix ledger: valid, retired, or unverified.
- **Investigate.** The agent runs diagnostics. The Fixer only applies a fix to a target seen in evidence or memory.
- **Remember.** A verified fix becomes a private memory, a postmortem artifact and a ledger row.
- **Share.** At the end of the night the datapack owner promotes valid fixes. The other shift finds them next time.
- **Forget on purpose.** When a remembered fix fails, the ledger retires it and the Fixer refuses it from then on.

Postgres keeps memory honest

Promoted memories cannot be edited or deleted over MCP. Retrieval alone cannot tell a good memory from a stale one. So the ledger decides.

```
INC-24 · checkout is throwing 502s, p99 latency 2s
```

```
apply_fix F01 pgb-checkout-rw on pgbouncer: REFUSED by Fixer: the fix ledger retired F01 pgb-checkout-rw for this alert (INC-20: F01 applied, symptoms per…
```

```
get_pool_stats on pgbouncer: pgbouncer: pool cl_waiting=0 cl_active=3 maxwait=0s
```

```
apply_fix F09 new_pricing_engine on checkout-api: disable_flag <flag name> new_pricing_engine applied on checkout-api: symptoms persist (checkout is throwing 50…
```

```
get_metrics on checkout-api: checkout-api: symptom confirmed (checkout is throwing 502s, p99 latency 2s); own cpu/mem normal; errors are on…
```

```
apply_fix F01 pgb-checkout-rw on pgbouncer: REFUSED by Fixer: the fix ledger retired F01 pgb-checkout-rw for this alert (INC-20: F01 applied, symptoms per…
```

```
get_queue on notify-queue: notify-queue: no queues owned or depth=0
```

```
get_logs on checkout-api: checkout-api: symptom confirmed (checkout is throwing 502s, p99 latency 2s); own cpu/mem normal; errors are on…
```

Same model, same 40 pages

qwen3:8b on a laptop. Left: whole runbook in the prompt. Right: Night Shift. Graded by a deterministic simulator.

	Pages	Runbook in prompt	Night Shift	Steps	Tokens per page
First seen	12	58%	**75%**	8.42 → 7.33	11,059 → 8,746
Repeat	25	52%	**72%**	8.88 → 5.48	11,962 → 7,460
Trap	3	33%	33%	9.67 → 9.33	13,328 → 13,853
All	40	52%	**70%**	8.8 → 6.33	11,793 → 8,325

On repeat pages Night Shift resolved 72% against 52%, with 38% fewer tokens. First-seen pages are the control: memory cannot help there yet.

Sharing is what does it

Same Night Shift agent with the collective layer switched off: each shift only sees its own private memories.

Repeat pages	Accuracy	Steps	Tokens per page
Private memory only	64%	7.52	7,909
Shared through Meko	**72%**	5.48	7,460

13 of the 25 repeats first happened on the other shift. Only the collective layer lets shift B use what shift A learned: +8 points on repeats.

Every Meko layer has a job

Conversations + messages	decision trace for every page, reasoning and plan
Token tracking	local-model tokens next to each trace
Private memory	memory_add after a verified fix, context_search at triage
Collective knowledge	runbook upload, memory_promote by the curator, knowledgebase_search
Artifacts	postmortem markdown per resolved page
Self-learning	shortfalls answered by a librarian, then replayed with selflearning_verify
Datapack sharing	shift B is a second Meko member with Contributor access
Postgres in the context	fix_validity ledger joined with every memory hit

Limits and next steps

- The incidents are simulated so that grading is exact. Real pages are messier.
- One 8B local model, no paid API. A larger model solves more first-seen pages in both modes.
- Traps stay hard: 33% of trap pages resolved, same as the runbook lane (33%). The ledger stops the old fix; finding the new cause is still up to the model.
- Next: feed real PagerDuty or Opsgenie pages, and let the ledger expire fixes after a deploy.

Code, replay and results: github.com/kangraemin/nightshift-meko